

```

this.BlockIndex = BlockIndex;
this.PreviousHashValue =
PreviousHashValue;
this.CurrentTime = CurrentTime;
this.DataBlock = DataBlock;
this.hash = this.calculateHash();
}
calculateHash()
{
return SHA256(this.BlockIndex + this.
PreviousHashValue + this.CurrentTime
+ JSON.stringify(this.DataBlock)).
toString();
}
}

```

The library of cryptography hash functions is invoked so that the dynamic hash can be generated for the blocks in the blockchain application. crypto-js can be used for the implementation of hash functions. The constructor declared in the class is used to initialise the values and these can be further called in the program. The hash of the block and transaction is processed with JSON to have the retrieved hash in the string.

Blockchain creation: The creation of a blockchain is required so that the blocks can participate in a secured environment and to validate transactions after successful authentication.

```

class SecuredBlockchain
{
// Genesis Block (The First Block of
Blockchain)
constructor()
{
this.chain = [this.
createGenesisSecuredBlock()];
}
createGenesisSecuredBlock()
{
return new SecuredBlock(0,
"01/01/2020", "Genesis SecuredBlock",
"0");
}

// Inserting Blocks
getLatestSecuredBlock()

```

```

{
return this.chain[this.chain.length -
1];
}
addSecuredBlock(newSecuredBlock) {
newSecuredBlock.PreviousHashValue =
this.getLatestSecuredBlock().hash;
newSecuredBlock.hash = newSecuredBlock.
calculateHash();
this.chain.push(newSecuredBlock);
}
// Validation and Authentication
Process
isChainValid()
{
for (let i = 1; i < this.chain.length;
i++)
{
const currentSecuredBlock = this.
chain[i];
const previousSecuredBlock = this.
chain[i - 1];
if (currentSecuredBlock.hash !==
currentSecuredBlock.calculateHash()) {
return false;
}
if (currentSecuredBlock.
PreviousHashValue !==
previousSecuredBlock.hash)
{
return false;
}
}
}

```

```

return true;
}
}

```

The genesis block refers to the very first block that was created in any blockchain network. There is no previous hash value of this block, as it is the initial block in the blockchain. After this genesis block, all other blocks are generated and inserted after validation using protocols and smart contracts. Once the blocks get activated in the blockchain, they are validated for further transactions and the transmission of data.

Implementing the blockchain is quite effective for e-governance, for which security and data privacy are very important. Many countries including Singapore, UAE, Japan, Switzerland, Australia and China are working on blockchain implementations for multiple applications. In developing countries like India, where data privacy is a serious matter, government services including Aadhar based citizen identification, biometric data storage, voter information, education data records, health records and many similar domains can be deployed using blockchain to enable high performance and security of data. **END** 

References

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